

DSA210

Project Proposal

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Analyzing The Factors That Makes a Song Popular on Spotify

In this project, I aim to explore what makes a song popular on Spotify by analyzing the relationship between different audio features and popularity scores. With the widespread use of music streaming platforms, understanding these patterns is both interesting and relevant. I will focus on features such as danceability, energy, tempo, and valence to see how they relate to a song's popularity, and whether they can be used to predict how successful a track might be.

The main dataset will be obtained from a publicly available Spotify dataset (e.g., Kaggle), which includes detailed audio features for a large number of songs. These features include variables such as danceability, energy, loudness, speechiness, acousticness, instrumentalness, tempo, and popularity. The data will be collected by downloading the dataset and then preprocessed by handling missing values and ensuring consistency across variables.

To enrich the analysis, I plan to incorporate additional information such as release year or artist-related attributes, which may help capture trends over time or differences between artists. The datasets will be merged using track or artist identifiers. After merging, songs with missing or inconsistent data will be filtered out. The final dataset is expected to include approximately 5,000–10,000 songs, and the target variable will be the popularity score of each track.